



WELCOME



KRITTER ASSIGNMENT

Village Economic Growth Intelligence

Identifying India's Top 100 Fastest Growing Districts
Using Satellite Data (2019 – 2024)

Tools: Google Earth Engine | Python | Google Colab

THE PROBLEM



No Real-Time GDP Data

Village & district-level GDP is not published or tracked in India in real-time



Outdated Census

Last Census published in 2011 — over 14 years of missing ground-level data



Surveys are Expensive

Traditional surveys are slow, costly, and cover only a fraction of India's 640,000+ villages

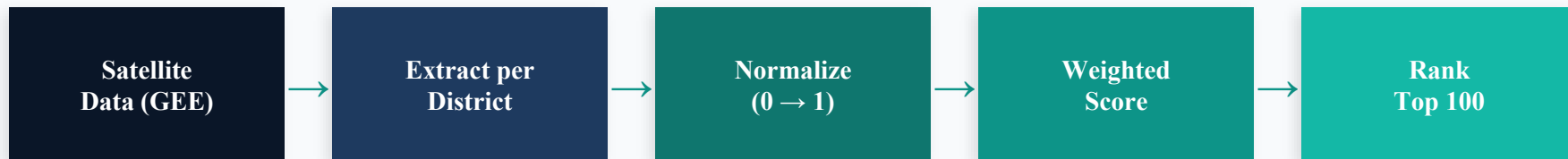
Solution: Satellite imagery captures economic signals -lights, construction, vegetation -continuously, for free, across all of India

DATA SOURCES & WHY WE CHOSE THEM

Signal	Dataset	Source	Why Used
Nighttime Lights	NOAA VIIRS DNB Monthly	Google Earth Engine	Best proxy for economic activity & electrification
Vegetation (NDVI)	MODIS MOD13A3	NASA / GEE	Measures agricultural productivity & land use change
Built-up Area	JRC GHSL Built Surface	EU Joint Research Centre	Tracks construction growth & urbanization
District Boundaries	FAO GAUL Level 2	FAO / GEE Built-in	640 Indian districts — base geographic unit

 Time Period: January 2019 → December 2024 (5-year change) | Platform: Google Earth Engine (free, browser-based)

METHODOLOGY & SCORING FRAMEWORK



$$\text{Growth Score} = (\text{NTL Change} \times 0.50) + (\text{NDVI Change} \times 0.25) + (\text{Built-up Change} \times 0.25)$$

50%

Nighttime Lights

Strongest proxy — directly reflects electricity use, commerce & activity

25%

NDVI Vegetation

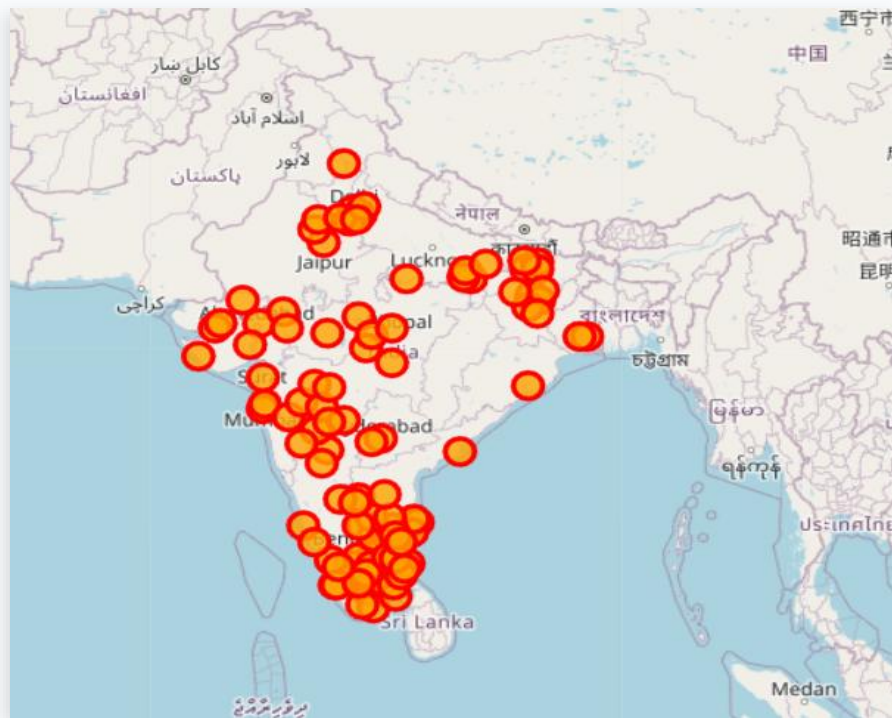
Agriculture is primary livelihood in rural India — crop health = income

25%

Built-up Area

Construction = investment flowing into a region — urbanization signal

KEY FINDINGS

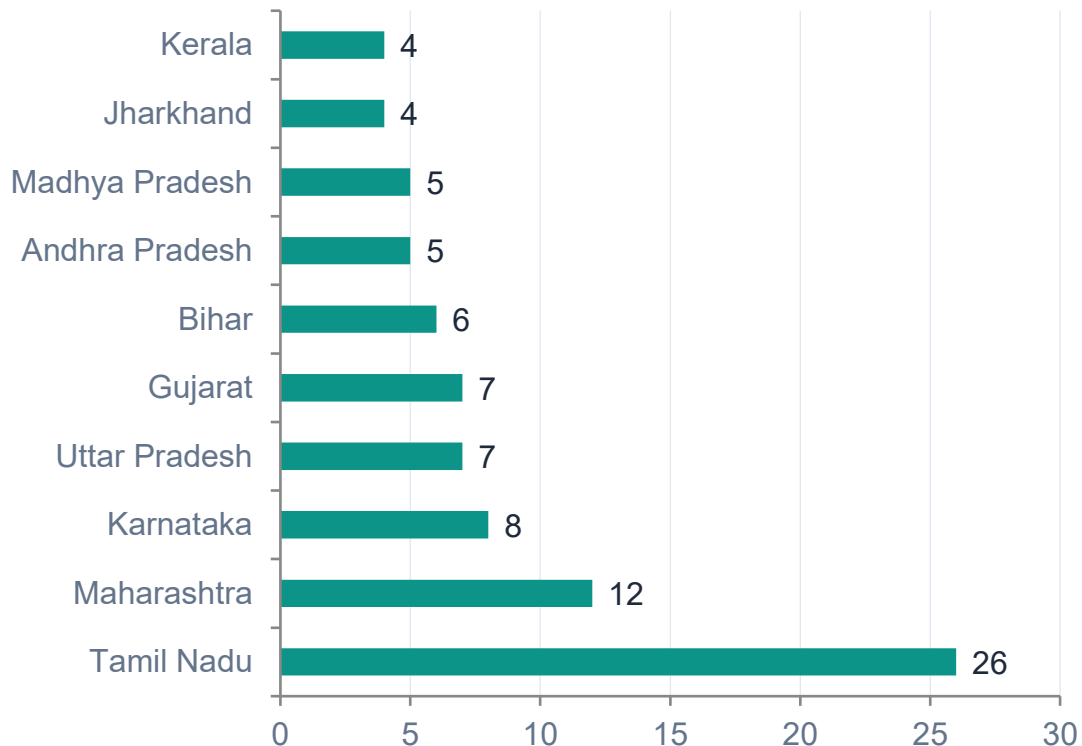


Most growth concentrated in Southern & Western India — Tamil Nadu, Maharashtra, and Karnataka lead

Top 10 Fastest Growing Districts

#	District	State	Score
1	Mumbai_City	Maharashtra	0.65
2	Daman	Daman & Diu	0.50
3	Chandigarh	Chandigarh	0.44
4	Mumbai Suburban	Maharashtra	0.39
5	Bangalore Urban	Karnataka	0.38
6	Gautam Buddha Nager	Uttar Pradesh	0.37
7	Salem	Tamil Nadu	0.365
8	Chennai	Tamil Nadu	0.362
9	Varanasi	Uttar Pradesh	0.33
10	Namakkal	Tamil Nadu	0.32

REGIONAL DISTRIBUTION OF GROWTH



Tamil Nadu

26 districts

Dominates top 100 — strong NTL & built-up growth



Maharashtra

12 districts

Mumbai corridor driving industrial growth



Karnataka

8 districts

Bengaluru tech hub radiating outward



Bihar & Jharkhand

10 districts

Rapid catch-up growth in lagging states

LIMITATIONS & NEXT STEPS

⚠ LIMITATIONS

- ⚠ District level used — village-level data not freely available
- ⚠ Proxy signals, not actual GDP or income measurements
- ⚠ Tamil Nadu has more districts — may inflate its ranking count
- ⚠ Cloud cover affects satellite readings in some regions
- ⚠ MODIS resolution (1 km) may miss small village-scale changes

✅ NEXT STEPS

- ✅ Obtain village-level boundaries (600,000+ villages)
- ✅ Add road network growth as an additional signal
- ✅ Add hospital & school density change indicators
- ✅ Validate findings with PMGSY & MNREGA ground data
- ✅ Use Machine Learning instead of manual weighted scoring

"With more time, a village-level ML pipeline could provide near real-time economic intelligence for all of India"

THANK YOU
